## 1. How does Seaborn relate to Matplotlib? (Strengths and weaknesses of each)

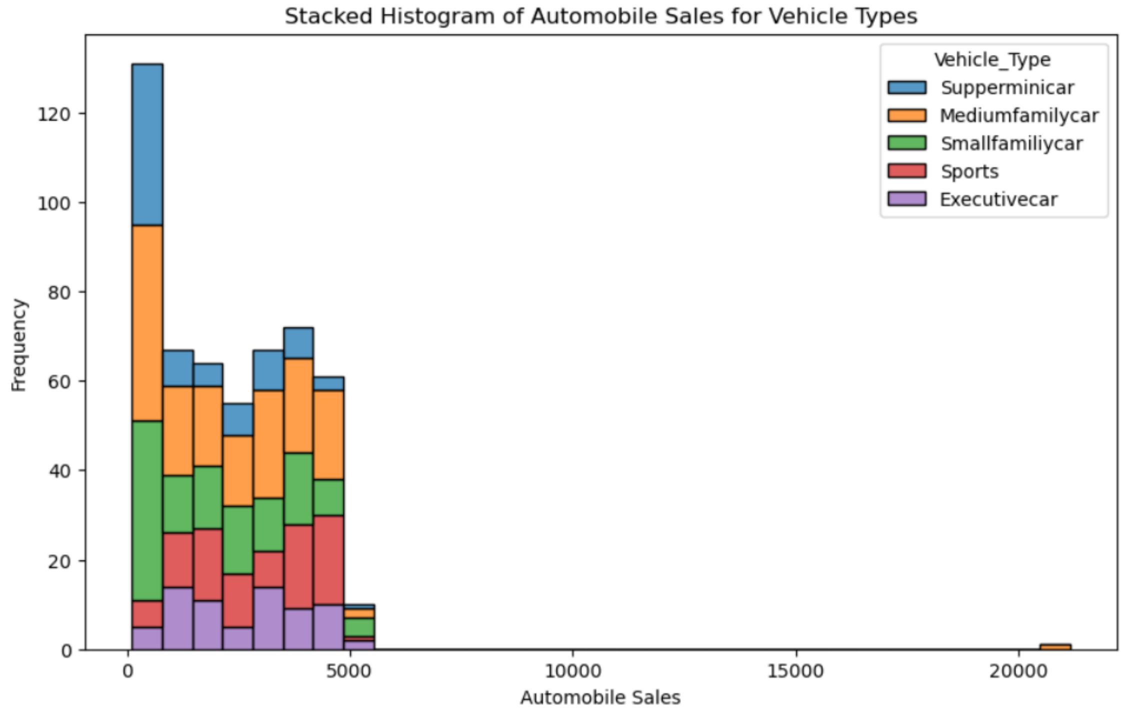
**Seaborn** is built on top of **Matplotlib** and provides a higher-level interface for creating more aesthetically pleasing and complex statistical graphics with fewer lines of code.

- **Strengths of Seaborn:**
  - **High-level API:** Seaborn makes it easier to create complex visualizations with just a few lines of code.
  - **Aesthetic default styles:** Seaborn automatically applies better aesthetics (colors, themes) for plots, whereas Matplotlib requires customization.
  - **Built-in statistical functions:** It simplifies the creation of statistical plots like regression lines, box plots, and violin plots.
  - **Better handling of Pandas DataFrames:** Seaborn works seamlessly with Pandas DataFrames, making it easier to plot data directly from them.
- **Weaknesses of Seaborn:**
  - **Less control:** While Seaborn simplifies plotting, it offers less control over every detail of the plot compared to Matplotlib.
  - **Not as versatile:** For very specific or complex customizations, Matplotlib is still the go-to library.
- **Strengths of Matplotlib:**
  - **Highly customizable:** Matplotlib allows you to customize every aspect of the plot (size, labels, colors, gridlines, etc.).
  - **Wide range of plots:** Matplotlib supports almost any kind of plot you can think of.
  - **Good for fine-grained control:** If you need to adjust specific details of a plot, Matplotlib gives you full control over the plot's appearance.
- **Weaknesses of Matplotlib:**
  - **Verbosity:** Matplotlib often requires more code to generate complex plots.
  - **Less intuitive for statistical plots:** It doesn't have as high-level API for statistical visualizations like Seaborn does.

## 2. Using Seaborn to Create a Stacked Histogram for Vehicle Sales by Type

We will first load the dataset and inspect the columns to identify the ones for sales and vehicle types

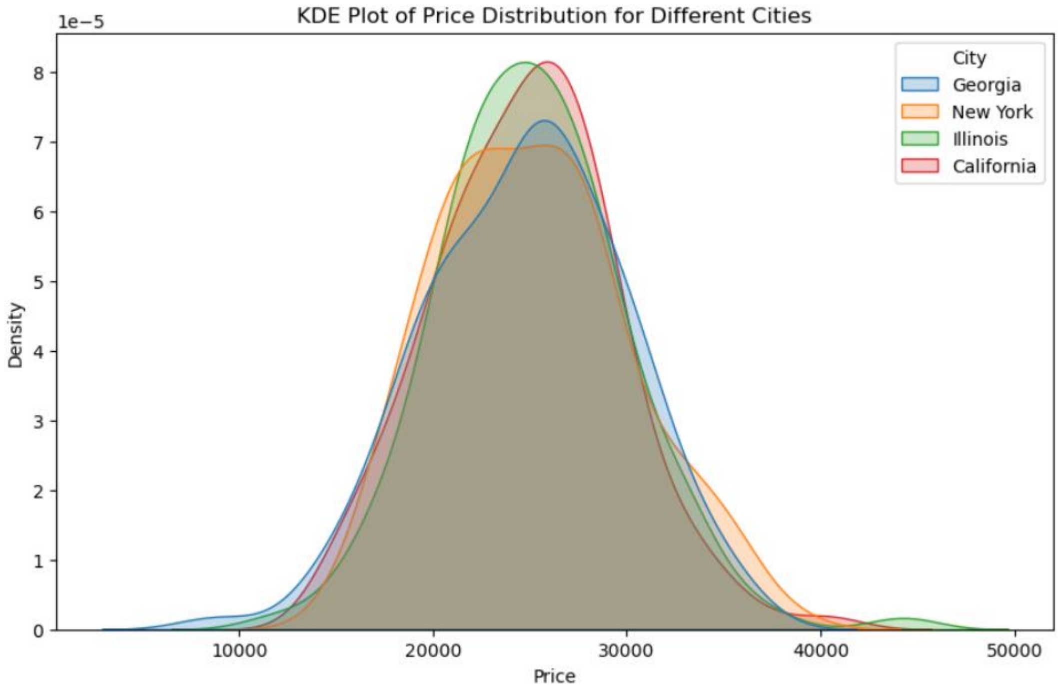
```python
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Load the dataset
url = "https://itv-contentbucket.s3.ap-south-1.amazonaws.com/Exams/AWP/Matplotlib/historical_automobile_sales.csv"
data = pd.read_csv(url)

# Inspect the first few rows to understand the structure
data.head()

# Create a stacked histogram to capture Automobile Sales for different Vehicle Types


plt.figure(figsize=(10, 6))
sns.histplot(data=data, x="Automobile_Sales", hue="Vehicle_Type", multiple="stack", kde=False)
plt.title('Stacked Histogram of Automobile Sales for Vehicle Types')
plt.xlabel('Automobile Sales')
plt.ylabel('Frequency')
plt.show()
```




## 2. KDE Plot for Price Distribution by City

We will create a KDE plot to visualize the distribution of `Price` for different `City`.

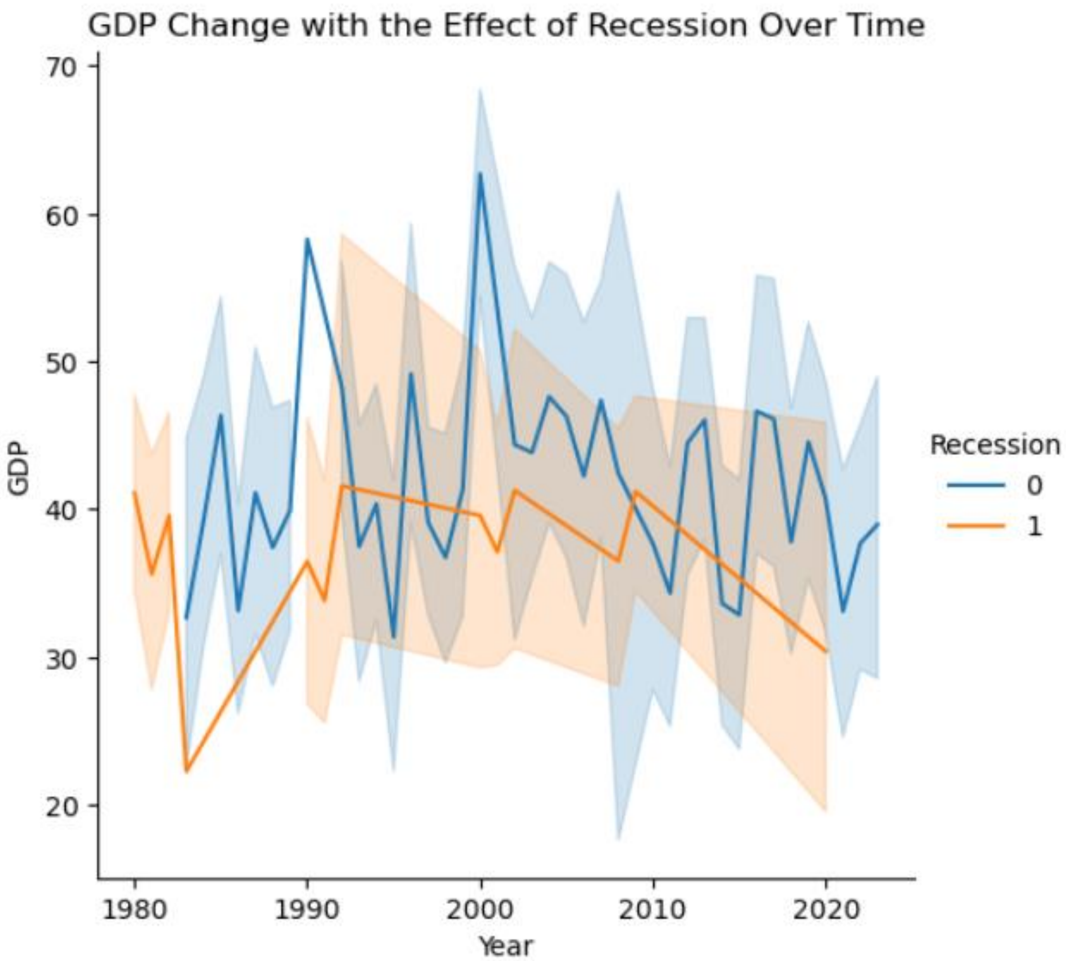
```python
plt.figure(figsize=(10, 6))
sns.kdeplot(data=data, x="Price", hue="City", common_norm=False, fill=True)
plt.title('KDE Plot of Price Distribution for Different Cities')
plt.xlabel('Price')
plt.ylabel('Density')
plt.show()
```




### 3. Relation Plot: GDP vs Year with the Effect of Recession

We will plot how `GDP` has changed over time with the effect of the `Recession`.

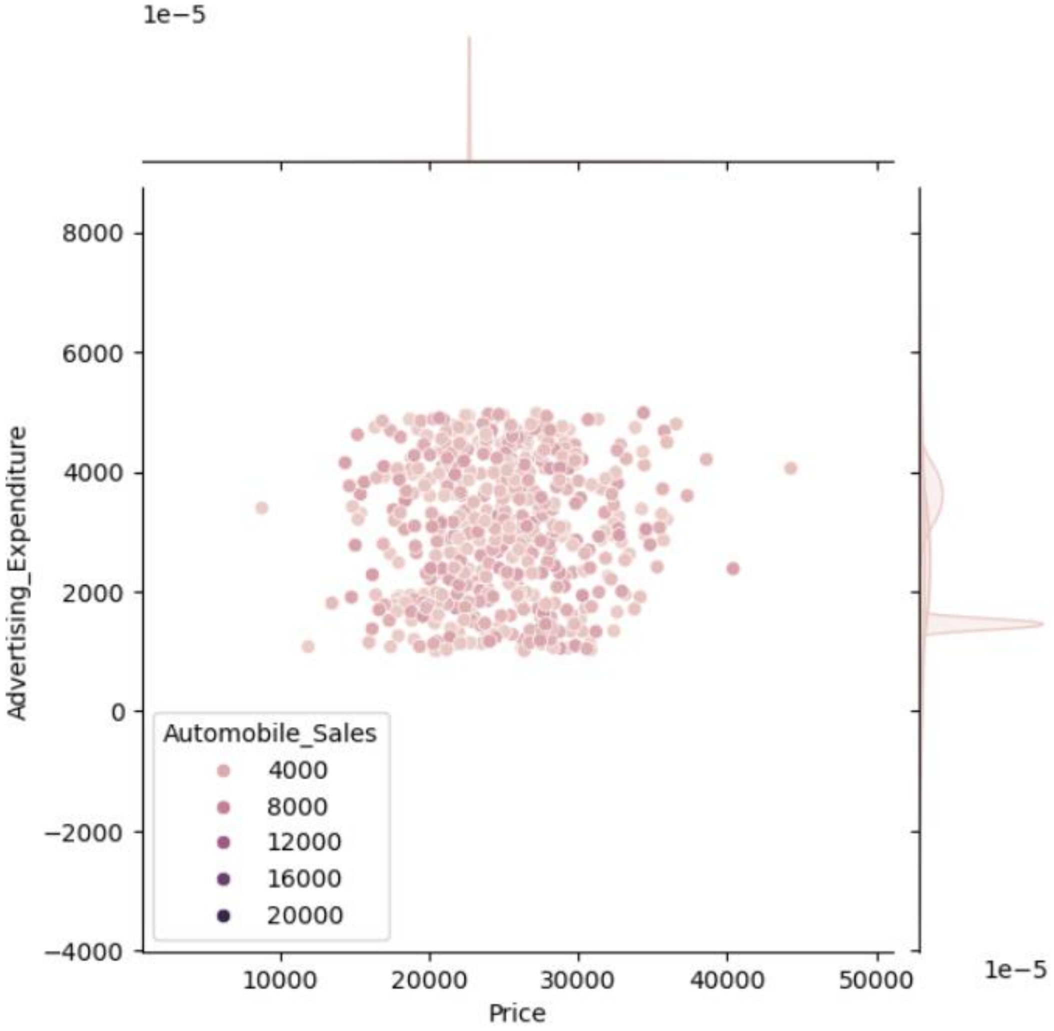
```
plt.figure(figsize=(10, 6))
sns.relplot(data=data, x="Year", y="GDP", hue="Recession", kind="line")
plt.title('GDP Change with the Effect of Recession Over Time')
plt.xlabel('Year')
plt.ylabel('GDP')
plt.show()
```




## 4. Joint Plot Between Price, Advertising Expenditure, and Automobile Sales

We will prepare a joint plot to visualize the relationship between `Price`, `Advertising_Expenditure`, and `Automobile_Sales`.

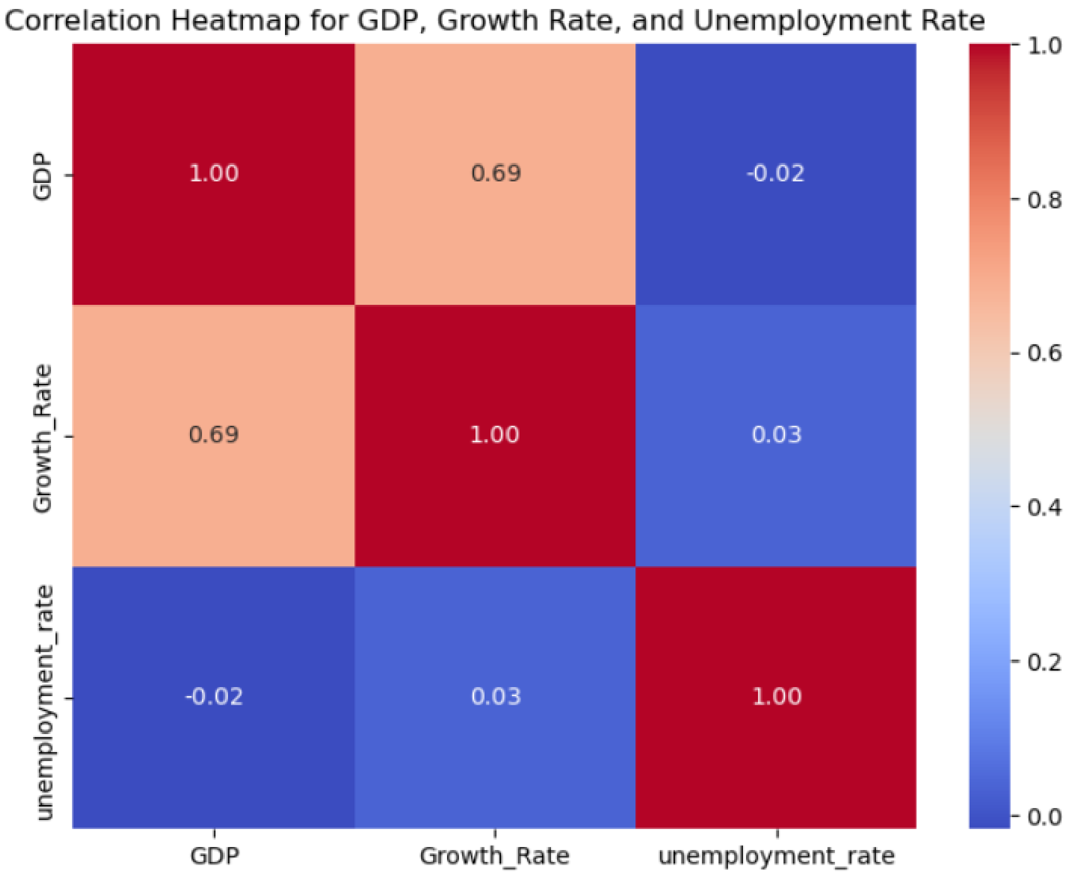
```
sns.jointplot(data=data, x="Price", y="Advertising_Expenditure", hue="Automobile_Sales", kind="scatter")
plt.show()
```




## 5. Heatmap of Correlation for GDP, Growth Rate, and Unemployment Rate

We will create a heatmap to show the correlation between `GDP`, `Growth_Rate`, and `Unemployment_Rate`.

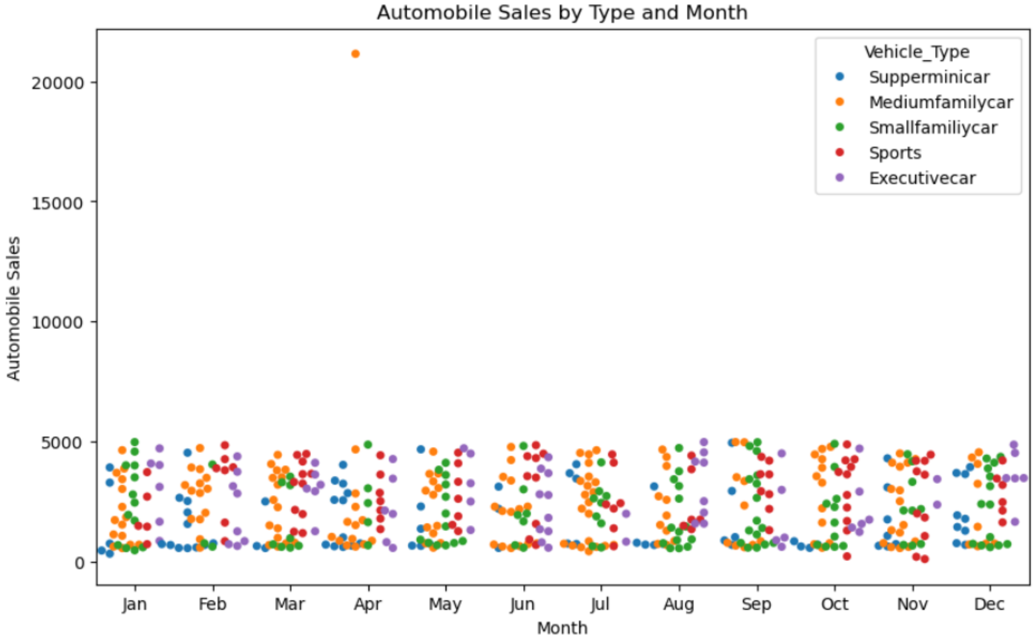
```
correlation_data = data[["GDP", "Growth_Rate", "unemployment_rate"]]
corr = correlation_data.corr()

plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap for GDP, Growth Rate, and Unemployment Rate')
plt.show()
```




## 6. Swarm Plot: Automobile Sales by Type and Month

We will use a swarm plot to visualize `Automobile_Sales` for each `Vehicle_Type` within each `Month`.

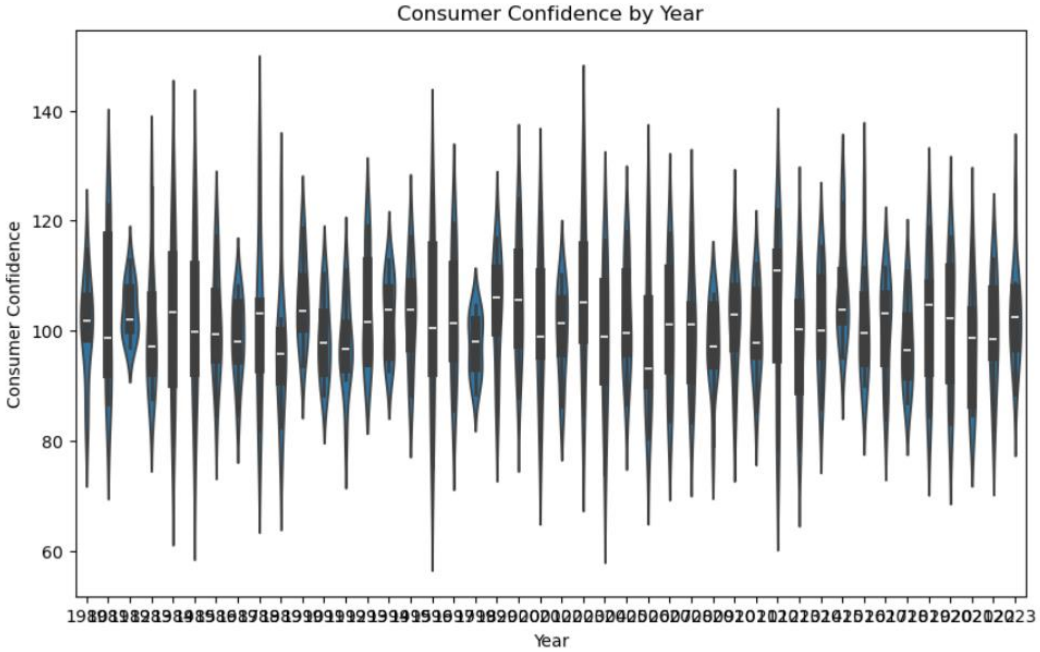
```
plt.figure(figsize=(10, 6))
sns.swarmplot(data=data, x="Month", y="Automobile_Sales", hue="Vehicle_Type", dodge=True)
plt.title('Automobile Sales by Type and Month')
plt.xlabel('Month')
plt.ylabel('Automobile Sales')
plt.show()
```




## 7. Violin Plot for Consumer Confidence by Year

We will create a violin plot to see how `Consumer_Confidence` has changed with each progressing `Year`.

```python
plt.figure(figsize=(10, 6))
sns.violinplot(data=data, x="Year", y="Consumer_Confidence")
plt.title('Consumer Confidence by Year')
plt.xlabel('Year')
plt.ylabel('Consumer Confidence')
plt.show()
```




## 8. FacetGrid of Competition and Advertising Expenditure

We will create a FacetGrid to view the histogram of `Advertising_Expenditure` for different `Competition` levels.

```python
g = sns.FacetGrid(data, col="Competition", col_wrap=4, height=4)
g.map(sns.histplot, "Advertising_Expenditure", kde=False)
plt.subplots_adjust(top=0.9)
g.fig.suptitle('Advertising Expenditure Distribution by Competition')
plt.show()
```